

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

ACID PROPERTIES & ISOLATION

2520030098

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Section – 3

Create database:

```
CREATE DATABASE IF NOT EXISTS bookflow10_db;
```

Output:

```
1 12:41:28 CREATE DATABASE IF NOT EXISTS bookflow10_db
```

Use database:

```
USE bookflow10_db;
```

Output:

```
2 12:50:54 USE bookflow10_db
```

Create Customer Table:

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    City VARCHAR(50)  
);
```

Output:

```
3 13:02:11 CREATE TABLE Customer ( Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL,... 0 rows
```

Create Account Table:

```
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output:

✓	4	13:03:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s)
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Create Transaction Table:

```
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output:

✓	5	13:04:09	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Acco	
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Insert Customer Data:

```

INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'Sushma', '9876543210', 'Hyderabad'),
(102, 'Sush', '9876543211', 'Vijayawada'),
(103, 'Sushuu', '9876543212', 'Bangalore'),
(104, 'Chuchuma', '9876543213', 'Chennai'),
(105, 'Sonu', '9876543214', 'Hyderabad');

```

Output:

```

6 13:05:54 INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Sushma', '9876543210', 'Hy...

```

Insert Account Data:

```

INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');

```

Output:

```

8 13:26:44 INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savi... 5 row

```

Insert Bank Transaction Data:

```

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);

```

Output:

```

9 13:27:56 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000),... 5 r

```

Display Data:

```
SELECT * FROM Customer;
```

Output:

	Customer_ID	Customer_Name	Phone	City
▶	101	Sushma	9876543210	Hyderabad
	102	Sush	9876543211	Vijayawada
	103	Sushuu	9876543212	Bangalore
	104	Chuchuma	9876543213	Chennai
	105	Sonu	9876543214	Hyderabad
*	NULL	NULL	NULL	NULL

```
SELECT * FROM Account;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Bank_Transaction;
```

Output:

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-16 13:27:56
	2	10002	DEPOSIT	15000.00	2026-08-16 13:27:56
	3	10003	WITHDRAW	20000.00	2026-08-16 13:27:56
	4	10004	DEPOSIT	5000.00	2026-08-16 13:27:56
	5	10005	WITHDRAW	10000.00	2026-08-16 13:27:56
*	NULL	NULL	NULL	NULL	NULL

Start A Transaction:

```
START TRANSACTION;
```

Output:

13 13:31:49 START TRANSACTION

Deposit Money:

```
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
```

Output:

14 13:33:05 UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001

Check Balance:

```
SELECT *
FROM Account
WHERE Account_No = 10001;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Commit Transaction:

```
COMMIT;
```

Output:

16 13:34:35 COMMIT

Withdraw Money:

```
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;
```

Output:

17 13:35:53 UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001

Check Balance:

```
SELECT *
FROM Account
WHERE Account_No = 10001;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL

Rollback:

```
ROLLBACK;
```

Output:

✔ 19 13:45:18 ROLLBACK

Start Transaction:

```
START TRANSACTION;
```

Output:

✔ 13 13:31:49 START TRANSACTION

First Update:

```
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
```

Output:

✔ 22 13:51:41 UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001

Create Savepoint:

```
SAVEPOINT Deposit1;
```

Output:

✔ 23 13:52:33 SAVEPOINT Deposit 1

Second Update:


```
UPDATE Account
SET Balance = Balance - 3000
WHERE Account_No = 10002;
```

Output:

```
✓ 24 14:00:28 UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002
```

Create Second Savepoint:

```
SAVEPOINT Withdrawal1;
```

Output:

```
✓ 25 14:02:17 SAVEPOINT Withdrawal1
```

Third Update:

```
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10003;
```

Output:

```
✓ 26 14:03:52 UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003
```

Rollback To Savepoint:

```
ROLLBACK TO SAVEPOINT Withdrawal1;
```

Commit:

```
COMMIT;
```

Output:

```
✓ 30 14:07:18 COMMIT
```

Check Accounts:

```
SELECT *
FROM Account;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	72000.00	Vijayawada
	10003	103	Current	130000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
✱	NULL	NULL	NULL	NULL	NULL